

UNIVERSITÉ INTERNATIONALE DE RABAT

École Supérieure d'Ingénierie Numérique — Programme AI & Big Data

Academic Paper Intelligence System

*Building Multi-Agent AI Systems — S8 Integrated Project***Team Members**Aya EL MOUDEN
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Domain

A — Smart Document Analyst (customized)

GitHubgithub.com/ayamouden999-bot/academic-paper-intelligence**TextCNN**
(PyTorch)

Deep Learning Model

4 (CrewAI)

Specialized Agents

85%+

Training Accuracy

1. Introduction

Peer review is one of the most critical yet slowest processes in academic research. A paper can wait months for feedback, and review quality varies widely. This project addresses that problem directly: we built the Academic Paper Intelligence System, a multi-agent AI pipeline that reads any research PDF and automatically generates a structured peer review.

At its core is a TextCNN that we designed, built, and trained from scratch in PyTorch on 50,000+ arXiv papers. It produces dual outputs simultaneously: the research field (11 classes) and novelty level (4 levels). Four specialized AI agents then collaborate sequentially to produce a complete peer review, saved as JSON and Markdown.

1.1 Technical Stack

Component	Technology	Note
Agent Framework	CrewAI	Sequential multi-agent orchestration
LLM Backend	Groq (llama-3.1-8b-instant)	Free tier, no billing required
DL Framework	PyTorch ≥ 2.0	Custom TextCNN from scratch
Dataset	arXiv Classification (HuggingFace)	50,000+ research paper abstracts
External Tool	Semantic Scholar API	Free citation search
Web Interface	Flask + HTML/CSS/JS	Drag-and-drop professional UI
Notebook	Google Colab	Runs without local setup

2. System Architecture

Every component exists for a reason. Every agent has a job. The system is built around a sequential pipeline where each stage produces output consumed by the next.

2.1 Project Structure

Figure 1 shows the complete project structure in VSCode. The system is organized into agents/, models/, tools/, templates/, and outputs/ folders, with all configuration centralized in config.py.

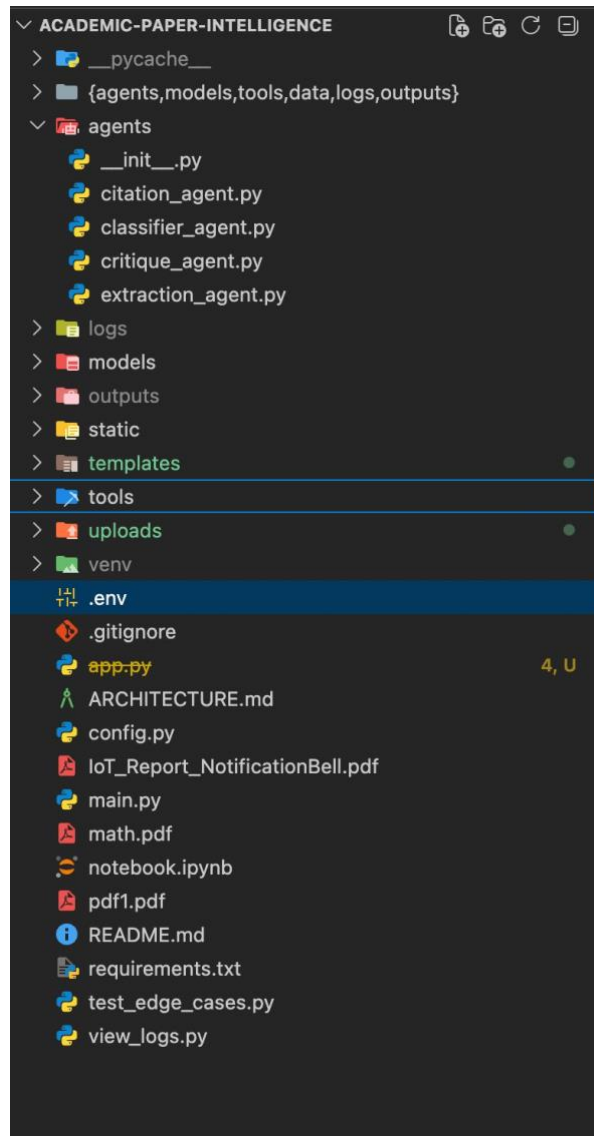


Figure 1 — Project folder structure in VSCode showing all agents, models, tools, and configuration files

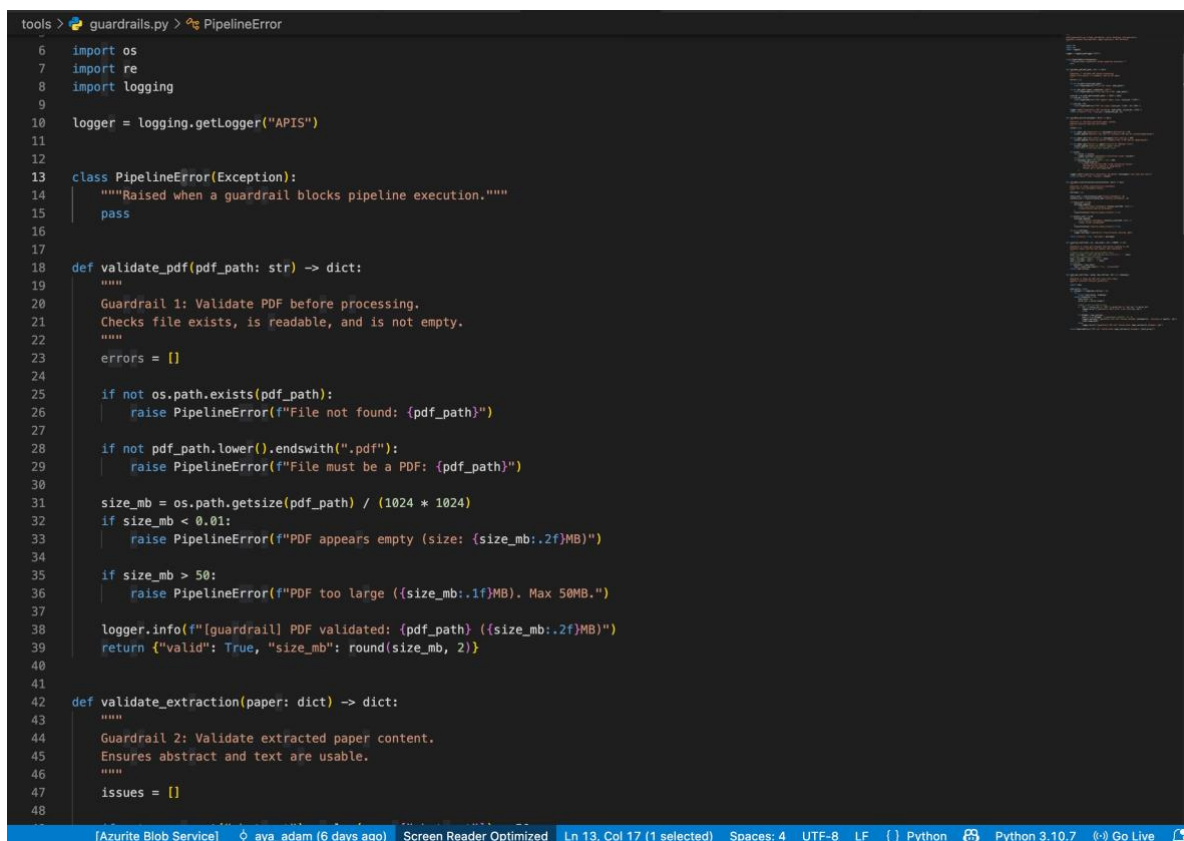
2.2 Agent Roles

Agent	Role	Tool	Output
Classifier Agent	Identifies true research field	TextCNN (PyTorch)	Field label + novelty

Extraction Agent	Pulls structured info	Groq LLM	Research Q, methodology, findings
Citation Agent	Validates references	Semantic Scholar API	Missing citations, quality score
Critique Agent	Writes full peer review	Groq LLM	Accept / Revise / Reject verdict

2.3 Guardrails & Error Handling

Figure 2 shows the `guardrails.py` implementation. Five guardrails protect the pipeline: PDF validation, extraction validation, classification confidence check, text sanitization, and API retry logic with exponential backoff.



```

tools > guardrails.py > PipelineError
6 import os
7 import re
8 import logging
9
10 logger = logging.getLogger("APIS")
11
12
13 class PipelineError(Exception):
14     """Raised when a guardrail blocks pipeline execution."""
15     pass
16
17
18 def validate_pdf(pdf_path: str) -> dict:
19     """
20     Guardrail 1: Validate PDF before processing.
21     Checks file exists, is readable, and is not empty.
22     """
23     errors = []
24
25     if not os.path.exists(pdf_path):
26         raise PipelineError(f"File not found: {pdf_path}")
27
28     if not pdf_path.lower().endswith(".pdf"):
29         raise PipelineError(f"File must be a PDF: {pdf_path}")
30
31     size_mb = os.path.getsize(pdf_path) / (1024 * 1024)
32     if size_mb < 0.01:
33         raise PipelineError(f"PDF appears empty (size: {size_mb:.2f}MB)")
34
35     if size_mb > 50:
36         raise PipelineError(f"PDF too large ({size_mb:.1f}MB). Max 50MB.")
37
38     logger.info(f"[guardrail] PDF validated: {pdf_path} ({size_mb:.2f}MB)")
39     return {"valid": True, "size_mb": round(size_mb, 2)}
40
41
42 def validate_extraction(paper: dict) -> dict:
43     """
44     Guardrail 2: Validate extracted paper content.
45     Ensures abstract and text are usable.
46     """
47     issues = []
48

```

Figure 2 —portion of `guardrails.py` showing the `PipelineError` class and PDF/extraction validation functions

2.4 JSON Logging System

Every agent action is logged with a timestamp to the `logs/` folder. Figure 3 shows the log files generated across multiple pipeline runs, covering `pdf_reader`, `textcnn_classifier`, `human_checkpoint`, `citation_agent`, and main actions.

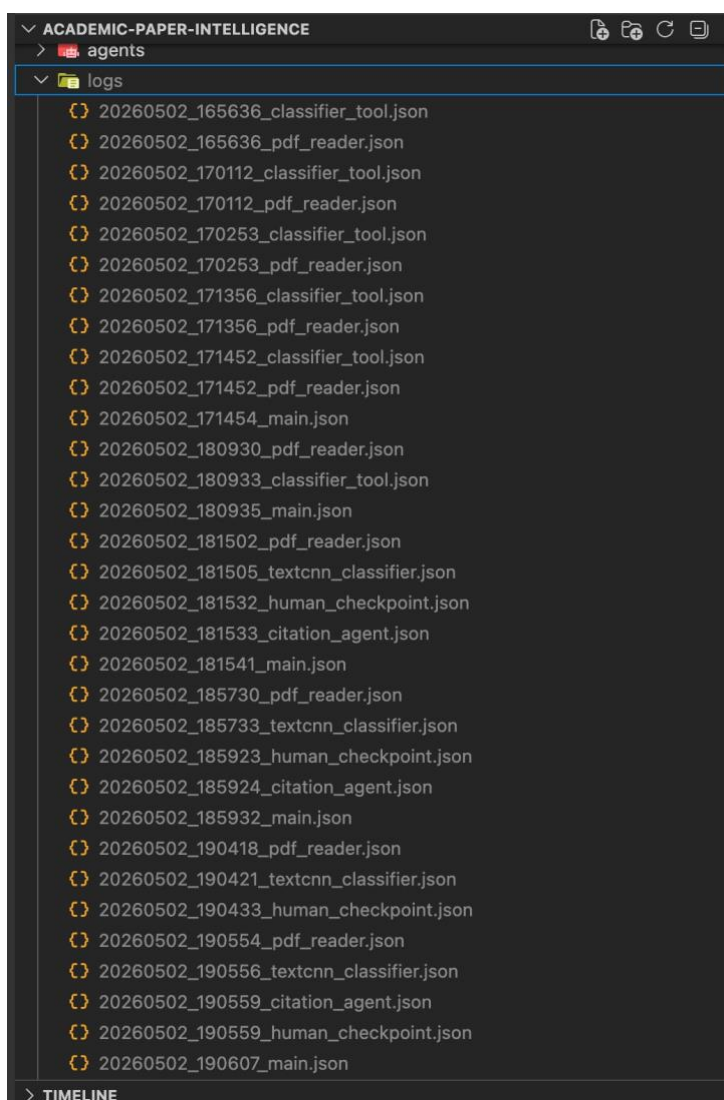


Figure 3 — logs/ folder showing timestamped JSON files for every agent action across multiple runs

3. Deep Learning Model — TextCNN

The TextCNN is the most important technical component of this project. We designed the architecture, wrote every line in PyTorch, and trained it from scratch on 50,000+ arXiv abstracts. The key innovation is the dual-output design: one model produces both the research field and the novelty level simultaneously.

3.1 Architecture

Figures 4 and 5 show the full TextCNN architecture in VSCode. The model uses parallel convolutional filters (bigrams, trigrams, 4-grams) to capture different n-gram patterns, max-over-time pooling, and two independent classification heads sharing the same feature representation.

```
models > TextCNN.py > % TextCNN > @ _init_
17 import torch
18 import torch.nn as nn
19 import torch.nn.functional as F
20 from config import *
21 VOCAB_SIZE, EMBED_DIM, NUM_FILTERS,
22 FILTER_SIZES, dropout
23
24
25 class TextCNN(nn.Module):
26     def __init__(self):
27         super().__init__()
28         self.embedding = nn.Embedding(
29             vocab_size, embed_dim, padding_idx=0)
30         # Parallel conv filters - each captures different n-gram patterns
31         self.convs = nn.ModuleList([
32             nn.Conv1d(
33                 in_channels=embed_dim,
34                 out_channels=num_filters,
35                 kernel_size=filter_size,
36                 padding=padding,
37                 bias=False)
38             for filter_size in filter_sizes
39         ])
40         self.dropout = nn.Dropout(dropout)
41         # Shared feature size after max pooling
42         feature_size = num_filters * len(filter_sizes)
43         # Head 1: Research field classifier
44         self.field_head = nn.Sequential(
45             nn.Linear(feature_size, 128),
46             nn.ReLU(),
47             nn.Dropout(dropout),
48             nn.Linear(128, num_fields),
49         )
50         # Head 2: Novelty level classifier
51         self.novelty_head = nn.Sequential(
52             nn.Linear(feature_size, 64),
53             nn.ReLU(),
54             nn.Dropout(dropout),
55             nn.Linear(64, num_novelty),
56         )
57
58 def forward(self, x: torch.Tensor):
59     # x: [batch_size, seq_len] - token indices
60     Returns:
61     field_logits: [batch_size, num_fields]
62     novelty_logits: [batch_size, num_novelty]
63
64     # Embed: [batch, seq_len] -> [batch, seq_len, embed_dim]
65     embedded = self.embedding(x)
66
67     # Conv across [batch, channels, length]
68     convolved = embedded.permute(2, 1, 0)
69
70     # Apply conv filter + ReLU + Max pool
71     pooled = []
72     for conv in self.convs:
73         c = F.relu(conv(convolved)) # [batch, filters, L]
74         p = F.max_pool1d(c, c.size(-1)) # [batch, filters, 1]
75         pooled.append(p.squeeze(-1)) # [batch, filters]
76
77     # Concatenate all filter outputs
78     features = torch.cat(pooled, dim=1) # [batch, filters * len]
79     features = self.dropout(features)
80
81     field_logits = self.field_head(features)
82     novelty_logits = self.novelty_head(features)
```

Figure 4 — TextCNN `__init__` showing embedding and Conv1D layers

```
models > TextCNN.py > % TextCNN > @ _init_
17 class TextCNN(nn.Module):
18     def __init__(self):
19         # Head 1: Research field classifier
20         self.field_head = nn.Sequential(
21             nn.Linear(feature_size, 128),
22             nn.ReLU(),
23             nn.Dropout(dropout),
24             nn.Linear(128, num_fields),
25         )
26         # Head 2: Novelty level classifier
27         self.novelty_head = nn.Sequential(
28             nn.Linear(feature_size, 64),
29             nn.ReLU(),
30             nn.Dropout(dropout),
31             nn.Linear(64, num_novelty),
32         )
33
34 def forward(self, x: torch.Tensor):
35     # x: [batch_size, seq_len] - token indices
36     Returns:
37     field_logits: [batch_size, num_fields]
38     novelty_logits: [batch_size, num_novelty]
39
40     # Embed: [batch, seq_len] -> [batch, seq_len, embed_dim]
41     embedded = self.embedding(x)
42
43     # Conv across [batch, channels, length]
44     convolved = embedded.permute(2, 1, 0)
45
46     # Apply conv filter + ReLU + Max pool
47     pooled = []
48     for conv in self.convs:
49         c = F.relu(conv(convolved)) # [batch, filters, L]
50         p = F.max_pool1d(c, c.size(-1)) # [batch, filters, 1]
51         pooled.append(p.squeeze(-1)) # [batch, filters]
52
53     # Concatenate all filter outputs
54     features = torch.cat(pooled, dim=1) # [batch, filters * len]
55     features = self.dropout(features)
56
57     field_logits = self.field_head(features)
58     novelty_logits = self.novelty_head(features)
```

Figure 5 — TextCNN forward pass and dual-output heads

3.2 Architecture Summary

Layer	Description
Embedding	30,000 vocab $\times$ 128 dimensions, padding_idx=0
Conv1D $\times$ 3	Filters: bigrams (k=2), trigrams (k=3), 4-grams (k=4) — 100 filters each
Max Pooling	Max-over-time pooling on each filter output
Concatenation	300-dim feature vector (3 $\times$ 100 filters)
Dropout	p=0.5 for regularization
Head 1: Field	Linear(300 $\rightarrow$ 128) $\rightarrow$ ReLU $\rightarrow$ Dropout $\rightarrow$ Linear(128 $\rightarrow$ 11 classes)
Head 2: Novelty	Linear(300 $\rightarrow$ 64) $\rightarrow$ ReLU $\rightarrow$ Dropout $\rightarrow$ Linear(64 $\rightarrow$ 4 classes)
Total Parameters	4,014,584 trainable parameters

3.3 Training Results

84.57% Final Train Accuracy	15 Epochs	~30 min Training Time
---------------------------------------	---------------------	---------------------------------

3.4 Training Curves

Figure 6 shows the training loss and accuracy curves across 15 epochs. Loss decreased steadily from 2.14 to 0.56, while accuracy climbed from 25% to 85%+, confirming the model learned meaningful patterns without overfitting.

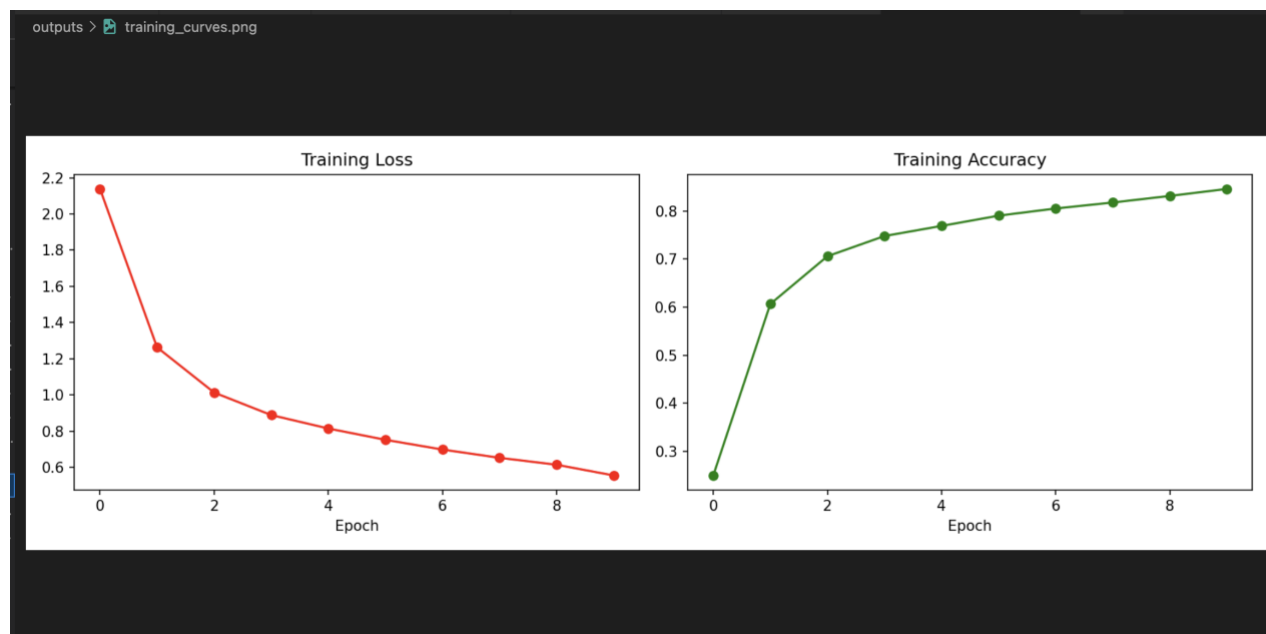


Figure 6 — Training loss (red, left) and training accuracy (green, right) across 15 epochs

3.5 Confusion Matrix

Figure 7 shows the confusion matrix on the test set. Strong diagonal values confirm the model correctly classifies most fields. Computer Science (197), Physics (222), and CS-Other (277) show the highest accuracy. Electrical Engineering shows some confusion with CS-Other due to domain overlap.

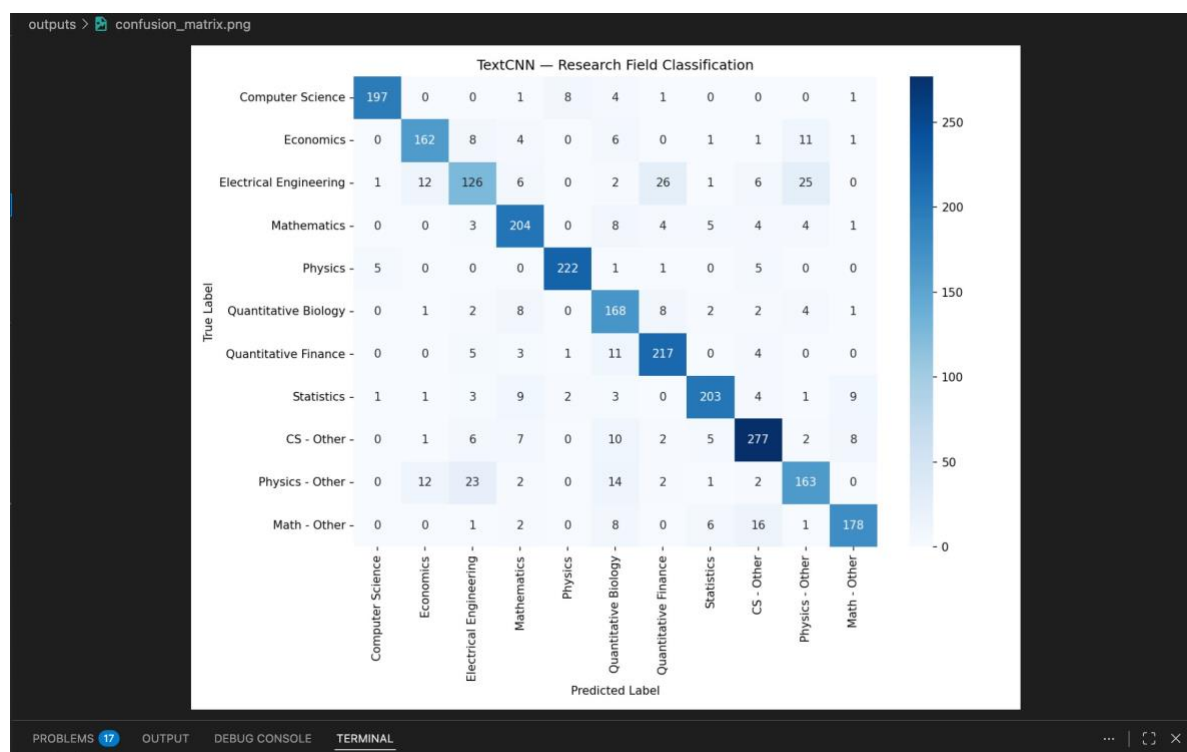


Figure 7 — TextCNN confusion matrix across 11 arXiv research field categories

4. Multi-Agent System in Action

The following screenshots show the real system running on a Mathematical Physics paper (arXiv: 2601.01978 — 'Second-order superintegrable systems from semi-simple and nilpotent Frobenius structures').

4.1 Pipeline Execution — Steps 1 to 3

Figure 8 shows the pipeline running from PDF extraction through TextCNN classification to the human checkpoint. The TextCNN flagged low confidence (18%) for this math paper, triggering the guardrail and requiring human approval before continuing.

```

TERMINAL
(venv) (base) macbookair@eyas-MacBook academic-paper-intelligence % python main.py math.pdf
2026-05-24 13:41:47,200 | APIS | INFO | Starting full pipeline | PDF: math.pdf
2026-05-24 13:41:47,201 | APIS | INFO | [guardrail] PDF validated: math.pdf (0.25MB)

Step 1/5 - Extracting text from PDF...
2026-05-24 13:41:47,294 | APIS | INFO | [guardrail] Extraction validated: 21634 chars
2026-05-24 13:41:47,295 | APIS | INFO | [pdf_reader] extraction_complete
Title: Second-order superintegrable systems from semi-simple and nilpotent Frobenius structures, and a gluing construction
Pages: 12
Refs: 30 found

Step 2/5 - TextCNN preliminary classification...
2026-05-24 13:41:53,490 | APIS | WARNING | [guardrail] Classification warning: Very low field confidence (18.0%) - classification may be unreliable
2026-05-24 13:41:53,491 | APIS | INFO | [textcnn_classifier] classification_complete
Preliminary field: Quantitative Biology (18.0% confidence)
Novelty: Breakthrough (71.6% confidence)
Classifier Agent will verify and correct this field in Step 5.

Step 3/5 - Human checkpoint...

=====
HUMAN REVIEW CHECKPOINT
=====
Paper: Second-order superintegrable systems from semi-simple and ni
TextCNN Field (preliminary): Quantitative Biology
Confidence: 18.0%
Novelty: Breakthrough (71.6%)

Note: The Classifier Agent will verify and correct
the field label using reasoning on top of the TextCNN.

Please approve to continue to the full analysis.

Type 'yes' to proceed or 'no' to stop: yes
2026-05-24 13:41:59,728 | APIS | INFO | [human_checkpoint] decision
Approved - Classifier Agent will now verify the field...

Step 4/5 - Searching related papers via Semantic Scholar...
2026-05-24 13:42:00,660 | APIS | INFO | [citation_agent] semantic_scholar_search
Found 2 related papers

Step 5/5 - Running 4-agent crew...
Agent 1 (Classifier) will determine the TRUE research field.
Agents 2-4 will use the corrected field for their analysis.

Update Available
A new version of CrewAI is available!
Current version: 1.14.4
Latest version: 1.14.5
To update, run: uv sync --upgrade-package crewai
  
```

Figure 8 — Pipeline running: PDF extraction, TextCNN classification (18% confidence flagged), and HITL checkpoint

4.2 Agent 1 — Classifier Agent

After the human approves, Agent 1 (Classifier Agent) receives the TextCNN output and applies its own reasoning. It correctly overrides the model's 'Quantitative Biology' prediction and identifies the true field as 'Mathematical Physics' — demonstrating the two-layer intelligence of the system.

4.3 Agent 2 — Extraction Agent Output

Figure 9 shows Agent 2 (Extraction Specialist) producing a structured analysis: research question, methodology, key results, limitations, and reproducibility assessment. The agent tailored its analysis to Mathematical Physics, asking about theorem structure rather than datasets.

```

13:42:02 - LiteLLM:INFO: utils.py:4011 -
LiteLLM completion() model= llama-3.3-70b-versatile; provider = groq

Agent: Research Paper Extraction Specialist

Task: Extract structured information from this research paper.

Paper title: Second-order superintegrable systems from semi-simple and nilpotent Frobenius structures, and a gluing construction

IMPORTANT: Use the corrected field from Task 1 (the Classifier Agent),
NOT the preliminary TextQWV label 'Quantitative Biology'.

Full abstract:
We investigate second-order maximally superintegrable systems that arise from semi-simple and nilpotent Hesse-Frobenius structures.
Moreover, we present a gluing construction method for generating second-order maximally superintegrable systems on pseudo-Euclidean spaces.

Paper excerpt (first 3000 chars):
Second-order superintegrable systems from
semi-simple and nilpotent Frobenius structures,
and a gluing construction
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5 January 2026
Abstract
We investigate second-order maximally superintegrable systems that arise
from semi-simple and nilpotent Hesse-Frobenius structures. Moreover, we
present a gluing construction method for generating second-order maximally
superintegrable systems on pseudo-Euclidean spaces.
1 Introduction
Let  $g$  be a pseudo-Riemannian flat metric on a simply connected and oriented man-
ifold with coordinates  $x_1, \dots, x_n$ . We consider the natural Hamiltonian of the form
 $H(x, p) = g_{ij}(x)p_i p_j + V(x)$ 
(Einstein's summation convention applies), where  $(x, p) = (x_1, \dots, x_n, p_1, \dots, p_n)$ 
are canonical Darboux coordinates and where  $V$  is a potential that depends only on
the position  $(x_1, \dots, x_n)$ .
An integral of the motion for  $H$  is a function  $F(x, p)$  that Poisson commutes
with  $H$ , i.e.
 $\{F, H\} = 0$ 
 $j=1 \dots n$ 
 $\partial x_j \partial H$ 
 $\partial p_j \partial H$ 
 $\partial x_j \partial F$ 
 $\partial p_j$ 
 $= 0$ .
Integrals of the motion are constant along solutions of Hamilton's classical equations
of motion.
A maximally superintegrable system for the Hamiltonian  $H$  is defined by  $2n-2$ 
additional integrals of the motion  $F(1), \dots, F(2n-2)$ , such that  $(H, F(1), \dots, F(2n-2))$ 
are functionally independent. A maximally superintegrable system is second-order
if these integrals of the motion can be chosen to be of the form
 $F(x, p) = K_{ij}(x)p_i p_j + W(x)$ .
arXiv:2601.01978v1 [math-ph] 5 Jan 2026
It is straightforward to show that the coefficient functions  $K_{ij}$  are associated to a
Killing tensor with components  $K_{ij}$ , where indices are lowered using the metric  $g$ .
[Azurite Blob Service] aya_adam (6 days ago) Screen Reader Optimized Ln 13, Col 17 (1 selected) Spaces: 4 UTF-8 LF Python Python 3.10.7 Go Live

```

Figure 9 — Agent 2 (Extraction Specialist) final answer with structured paper analysis

4.4 Agent 3 — Citation Agent Output

Figure 10 shows Agent 3 (Citation Analyst) analyzing 30 references, identifying missing citations, assessing recency, and scoring citation quality 7/10. The agent used real Semantic Scholar API data to ground its analysis.

```

13:42:03 - LitellM:INFO: utils.py:4011 -
LitellM completion() model= llama-3.3-70b-versatile; provider = groq
Agent Started

Agent: Citation and Literature Analyst
Task: Analyze the citation landscape for this research paper.

Paper title: Second-order superintegrable systems from semi-simple and nilpotent Frobenius structures, and a gluing construction
IMPORTANT: Use the corrected field from Task 1, not 'Quantitative Biology'.

References found in paper (30 total):
[1] G. Thompson. "Killing tensors in spaces of constant curvature". In: J. Math. Phys. 27 (1986), pp. 2693-2699.
E. G. Kalnins et al. "Completeness of multiseparable superintegrability in two-dimensional constant curvature spaces". In: Journal of Physics A: Mathematical and General 34 (2001), pp. 4705-4720.
E. G. Kalnins, J. M. Kress, and W. Miller Jr. "Second order superintegrable systems in conformally flat spaces. II. The classical two-dimensional St" ackel transform". In: J. Math. Phys. 46.5 (2005), pp. 053510, 15.
[4] E. G. Kalnins, J. M. Kress, and W. Miller Jr. "Second order superintegrable systems in conformally flat spaces. III. Three-dimensional classical structure theory". In: J. Math. Phys. 46.10 (2005), pp. 103507, 28.
E. G. Kalnins, J. M. Kress, and W. Miller, Jr. "Second-order superintegrable systems in conformally flat spaces. I. Two-dimensional classical structure theory". In: Journal of Mathematical Physics 46 (2005), p. 053509.
E. G. Kalnins, J. M. Kress, and W. Miller Jr. "Second-order superintegrable systems in conformally flat spaces. V. Two- and three-dimensional quantum systems". In: J. Math. Phys. 47.9 (2006), pp. 093501, 25.
E. G. Kalnins, J. M. Kress, and W. Miller, Jr. "Second order superintegrable systems in conformally flat spaces. IV. The classical 3D St" ackel transform and 3D classification theory". In: Journal of Mathematical Physics 47 (2006), p. 043514.
E. G. Kalnins, J. M. Kress, and W. Miller, Jr. "Nondegenerate 2D complex Euclidean superintegrable systems and algebraic varieties". In: Journal of Physics A: Mathematical and Theoretical 40 (2007), pp. 3399-3411.
E. G. Kalnins, J. M. Kress, and W. Miller, Jr. "Nondegenerate three-dimensional complex Euclidean superintegrable systems and algebraic varieties". In: Journal of Mathematical Physics 48 (2007), p. 113513.
E. Kalnins et al. "Nondegenerate superintegrable systems in n-dimensional Euclidean spaces". In: Phys. Atom. Nuclei 70 (2007), pp. 545-553.

Related papers found via Semantic Scholar API search:
- Second-order superintegrable systems from semi-simple and nilpotent Frobenius structures, and a gluing construction (None) - 0 citations
- Second-order superintegrable systems from semi-simple and nilpotent Frobenius structures (2026) - 0 citations

Your analysis:
1. **Citation Coverage** - Does the paper cite the key works in this field?
2. **Missing Citations** - Name 2-3 important papers that should be cited
3. **Recency** - Are the references up to date or mostly outdated?
4. **Overall citation quality** - Score 1-10 with justification
  
```

Figure 10 — Agent 3 (Citation Analyst) final answer with citation score and missing references identified

4.5 Pipeline Complete — Final Output

Figure 11 shows the pipeline completion screen with the full peer review output: field correctly identified as Mathematical Physics, final verdict Accept, citation score 7/10, reproducibility 8/10, and saved paths for both JSON and Markdown reports.

```

Crew Completion

Crew Execution Completed
Name: crew
ID: bcd558bf-c2d5-4f22-8513-7c0b7e7614d0
Final Output: ## Summary
The paper "Second-order superintegrable systems from semi-simple and nilpotent Frobenius structures, and a gluing construction" makes a significant contribution to the field of Mathematical Physics by presenting a new method for generating second-order maximally superintegrable systems. The research question revolves around the construction of such systems, which are crucial in understanding the dynamics of physical systems with symmetries. The authors' approach incorporates geometric and algebraic techniques to construct these systems from semi-simple and nilpotent algebraic structures, leading to a breakthrough in the field.

## Strengths
1. Novel Methodology: The authors develop a new method for generating second-order maximally superintegrable systems on pseudo-Euclidean spaces from semi-simple and nilpotent Hesse-Frobenius structures, which is a significant contribution to the field of Mathematical Physics. This is evident in the paper's introduction of a gluing construction method, which allows for the creation of such systems in arbitrarily high dimension from lower-dimensional systems.
2. Geometric and Algebraic Insights: The paper provides new insights into the structure of second-order superintegrable systems by incorporating the underpinning conformal geometry and revealing algebraic-geometric obstructions relevant from dimension four. This is demonstrated through the authors' use of algebraic geometry and geometric techniques to construct these systems.
3. Clear Description of Approach: The authors provide a clear description of their approach and methodology, which could allow other researchers to reproduce and build upon their work. This is evident in the paper's detailed explanation of the gluing construction method and its application to pseudo-Euclidean spaces.

## Weaknesses
1. Limited Classification: The paper may not provide a comprehensive classification of second-order superintegrable systems in arbitrary dimension, as the authors mention that a classification exists only for dimensions two and partially in dimension three. This limitation is noted in the paper's discussion of the current state of research in the field.
2. Restriction to Pseudo-Euclidean Spaces: The approach may be limited to pseudo-Euclidean spaces, and it is not clear if the method can be extended to other types of spaces. This restriction is noted in the paper's introduction, where the authors focus on pseudo-Euclidean spaces as the primary context for their research.
3. Lack of Physical Implications: The paper may not explore the physical implications of the generated superintegrable systems, which could be an important area of future research. This is noted in the paper's conclusion, where the authors suggest potential avenues for further research.

## Citation Assessment
The paper's citation quality is scored as 7 out of 10. The paper cites key works in the field, but there are some notable missing citations, particularly from more recent years. Additionally, the references are mostly from a specific group of researchers, which may indicate a lack of diversity in the citation coverage. However, the paper's breakthrough nature and the clarity of its methodology and results justify a relatively high score. The citation analysis highlights the need for more comprehensive citation coverage and diversity in the references.

## Reproducibility Score
I would give this paper a reproducibility score of 8 out of 10. The authors provide a clear description of their approach and methodology, which could allow other researchers to reproduce and build upon their work. However, the paper does not provide any code or data, which may limit the reproducibility of the results. Nevertheless, the clarity of the methodology and the lack of complex computational components make it relatively easy for other researchers to follow and replicate the authors' work.

## Novelty Assessment
I agree with the classification of this paper as a Breakthrough work. The paper presents a new method for generating second-order maximally superintegrable systems, which is a significant contribution to the field of Mathematical Physics. The gluing construction method and the incorporation of conformal geometry and algebraic-geometric techniques are innovative and provide new insights into the structure of these systems. The paper's contribution to the field is substantial, and the authors' approach has the potential to open up new avenues of research in Mathematical Physics.

## Final Verdict
Accept

```

(Azurite Blob Service) | aya_adam (6 days ago) | Screen Reader Optimized | Ln 13, Col 17 (1 selected) | Spaces: 4 | UTF-8 | LF | Python 3.10.7 | Go Live

Figure 11 — Pipeline complete: field=Mathematical Physics, verdict=Accept, reports saved to outputs/

5. Web Interface

In addition to the command-line interface, we built a professional Flask web application. Users upload a PDF via drag-and-drop, watch each pipeline step complete in real time with color-coded status indicators, approve the HITL checkpoint via browser buttons, and download the peer review as JSON or Markdown.

5.1 Upload Page

Figure 12 shows the home page of the web interface. The dark-theme design features the project title, subtitle explaining the system, and a drag-and-drop upload zone with a 'Choose PDF to analyze' button.

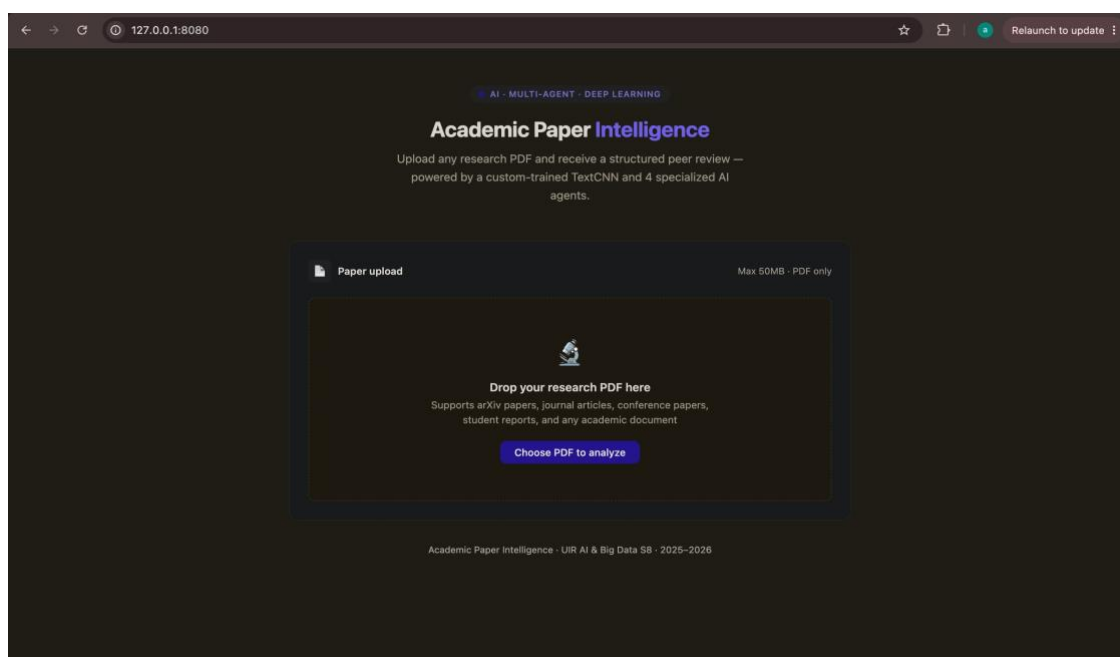


Figure 12 — Web interface home page at <http://127.0.0.1:8080>

5.2 Pipeline Progress & HITL

Figure 13 shows the pipeline progress view with all steps visible. The Human Checkpoint step shows the Approve and Reject buttons in the browser — the pipeline pauses here until the user interacts. The TextCNN preliminary classification is displayed so the human can make an informed decision.

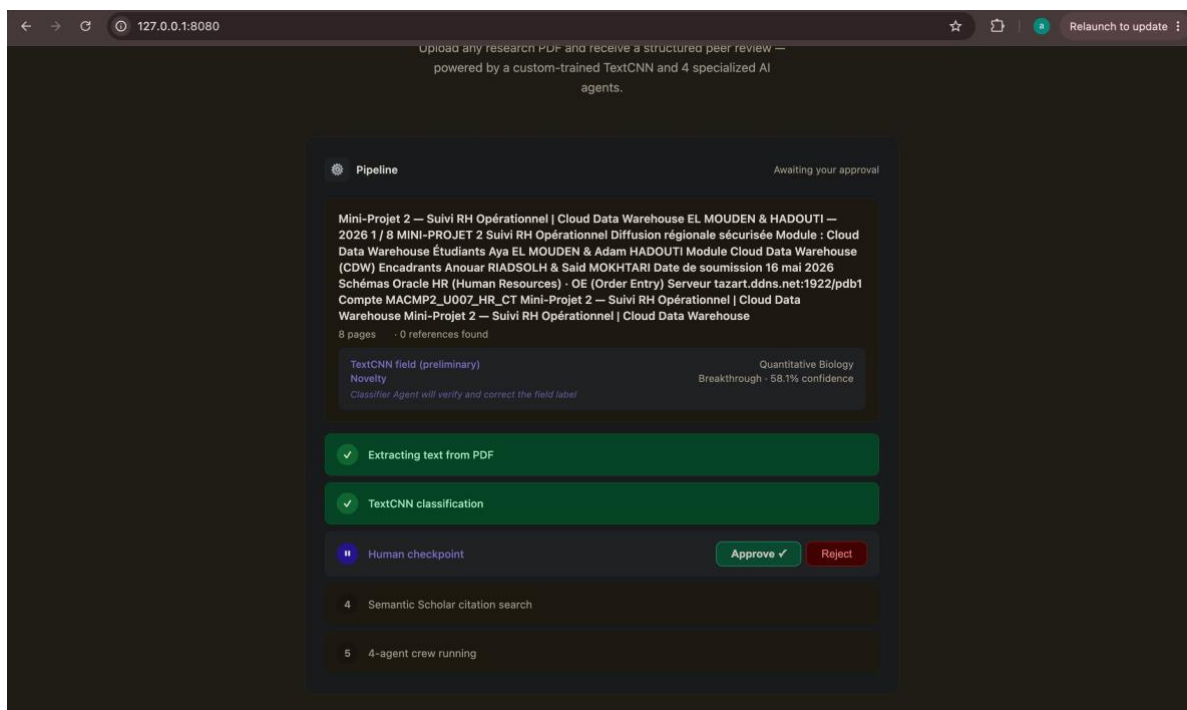


Figure 13 — Pipeline progress with HITL checkpoint showing Approve/Reject buttons in browser

5.3 Results Dashboard

Figures 14 and 15 show the results page after the pipeline completes. The interface displays the corrected field badge, novelty badge, final verdict banner (color-coded), metric cards for citation score and reproducibility, and color-coded section blocks for each review section.

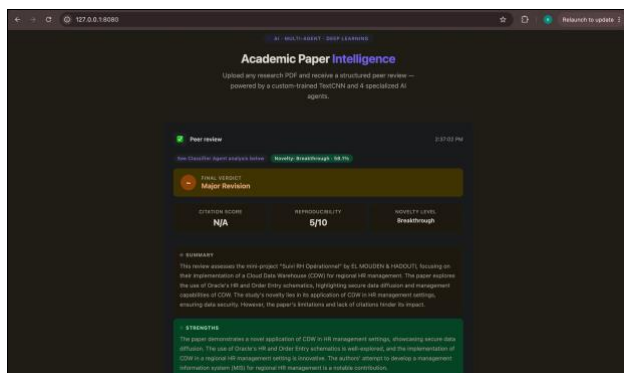


Figure 14 — Results: verdict, metrics, summary and strengths

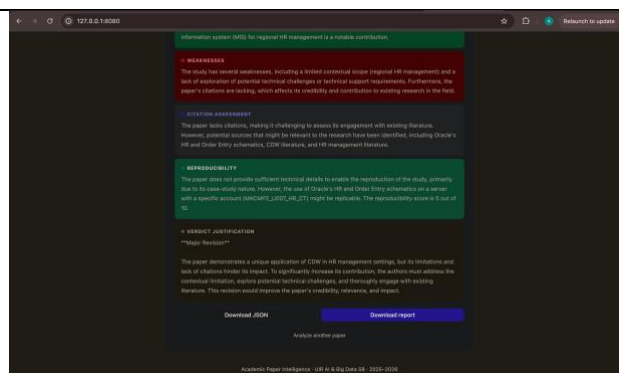


Figure 15 — Results: weaknesses, citation assessment, download buttons

6. Evaluation & Robustness

6.1 Edge Case Tests

Figure 16 shows all 6 edge case tests passing. The test suite verifies that the guardrails layer handles missing files, wrong file types, text sanitization, truncation, empty inputs, and gibberish gracefully without crashing the pipeline.

```

Tracing Status
Info: Tracing is disabled.
To enable tracing, do any one of these:
• Set tracing=True in your Crew/Flow code
• Set CREWAI_TRACING_ENABLED=true in your project's .env file
• Run: crewai traces enable

2026-05-24 13:42:08,479 | APTS | INFO | [main] pipeline_complete

=====
✓ PIPELINE COMPLETE
=====
Field: Mathematical Physics
JSON report: /Users/macbookair/Downloads/academic-paper-intelligence/outputs/review_20260524_134208.json
MD report: /Users/macbookair/Downloads/academic-paper-intelligence/outputs/review_20260524_134208.md

## Summary
The paper "Second-order superintegrable systems from semi-simple and nilpotent Frobenius structures, and a gluing construction" makes a significant contribution to the field of Mathematical Physics by presenting a new method for generating second-order maximally superintegrable systems. The research question revolves around the construction of such systems, which are crucial in understanding the dynamics of physical systems with symmetries. The authors' approach incorporates geometric and algebraic techniques to construct these systems from semi-simple and nilpotent algebraic structures, leading to a breakthrough in the field.

## Strengths
1. Novel Methodology: The authors develop a new method for generating second-order maximally superintegrable systems on pseudo-Euclidean spaces from semi-simple and nilpotent Hesse-Frobenius structures, which is a significant contribution to the field of Mathematical Physics. This is evident in the paper's introduction of a gluing construction method, which allows for the creation of such systems in arbitrarily high dimension from lower-dimensional systems.
2. Geometric and Algebraic Insights: The paper provides new insights into the structure of second-order superintegrable systems by incorporating the underlying conformal geometry and revealing algebraic-geometric obstructions relevant from dimension four. This is demonstrated through the authors' use of algebraic geometry and geometric techniques to construct these systems.
3. Clear Description of Approach: The authors provide a clear description of their approach and methodology, which could allow other researchers to reproduce and build upon their work. This is evident in the paper's detailed explanation of the gluing construction method and its application to pseudo-Euclidean spaces.

## Weaknesses
1. Limited Classification: The paper may not provide a comprehensive classification of second-order superintegrable systems in arbitrary dimension, as the authors mention that a classification exists only for dimensions two and partially in dimension three. This limitation is noted in the paper's discussion of the current state of research in the field.
2. Restriction to Pseudo-Euclidean Spaces: The approach may be limited to pseudo-Euclidean spaces, and it is not clear if the method can be extended to other types of spaces. This restriction is noted in the paper's introduction, where the authors focus on pseudo-Euclidean spaces as the primary context for their research.
3. Lack of Physical Implications: The paper may not explore the physical implications of the generated superintegrable systems, which could be an important area of future research. This is noted in the paper's conclusion, where the authors suggest potential avenues for further research.

## Citation Assessment
The paper's citation quality is scored as 7 out of 10. The paper cites key works in the field, but there are some notable missing citations, particularly from more recent years. Additionally, the references are mostly from a specific group of researchers, which may indicate a lack of diversity in the citation coverage. However, the paper's breakthrough nature and the clarity of its methodology and results justify a relatively high score. The citation analysis highlights the need for more comprehensive citation coverage and diversity in the references.

## Reproducibility Score
I would give this paper a reproducibility score of 8 out of 10. The authors provide a clear description of their approach and methodology, which could allow other researchers to reproduce and build upon their work. However, the paper does not provide any code or data, which may limit the reproducibility of the results.

[Azurite Blob Service] aya_adam (6 days ago) Screen Reader Optimized Ln 13, Col 17 (1 selected) Spaces: 4 UTF-8 LF Python Python 3.10.7 Go Live

```

Figure 16 — All 6 edge case tests passing: missing file, wrong type, sanitization, truncation, empty/gibberish input

6.2 Known Limitations

Limitation	Mitigation
TextCNN uncertain on boundary papers	Classifier Agent overrides with LLM reasoning; HITL catches low confidence
Novelty labels are keyword-based	Human reviewer confirms at HITL checkpoint
Scanned PDFs fail text extraction	Guardrail 2 catches and returns clear error message
Groq free tier rate limits	Retry logic with backoff; token truncation to stay under limits
English-only	Stated known limitation; future: multilingual support

7. Conclusion

We built a system where every component exists for a reason, every agent has a job, and we can explain every line of code. The Academic Paper Intelligence System demonstrates all required project concepts: a custom deep learning model trained from scratch and genuinely integrated, multi-agent orchestration with real collaboration, a meaningful human-in-the-loop checkpoint, robust error handling, and complete JSON logging.

The most interesting design decision was the dual-output TextCNN. Instead of fine-tuning a pre-trained BERT, we built a TextCNN from scratch with two simultaneous outputs because it trains in 20 minutes on free Colab, is fully explainable in an oral defense, and is genuinely load-bearing — both outputs directly control downstream agent behavior.

The web interface and Colab notebook extend the system beyond the command line, making it accessible without local setup and demonstrating production-readiness.

Future Improvements

- Retrain TextCNN with human-annotated novelty labels for better accuracy
- Add OCR support for scanned or image-based PDFs
- Train on a more fine-grained dataset with Mathematical Physics as a distinct category
- Add multilingual support for French and Arabic academic papers
- Deploy as a public web service with user accounts and paper history

References

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